Sparse Bayesian Model for Canonical factorization of multiblock data

---

1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data structures commonly arise when multiple sets of variables are collected on the same individuals to capture different psychological domains. For example, distinct blocks may include self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

The four expected objectives of exploratory multiblock data analysis are: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. An unified algebraic framework based on factorization strategies, denoted *canonical factorisation*, was proposed by [Bougeard et al., 2024] for multiblock methods to achieve these objectives. In practice, canonical factorisation provides a structured framework for analysing data variability at multiple levels: differences between and within blocks, differences between observations by projecting them onto a common subspace. It also facilitates the harmonisation of outputs produced by different methods and supports a more objective criterion-based selection among competing approaches.

2 Canonical factorisation

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals.
- $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks.
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$.
- $\Sigma_k = diag(\sigma_k^{(1)}, ..., \sigma_k^{(R)})$ are diagonal matrices
- $\|.\|$ correspond to the Frobenius norm.

Consider N individuals characterized by K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables. Let $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks. Without loss of generality, the blocks are assumed to be column-centered.

The $(K+1)-$blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K, \mathbf{X})$ can be factorized in dimension $R = rank(\mathbf{X})$ as follows (see [Bougeard et al., 2024] for the demonstration)

$$\begin{cases} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal \quad \text{for} \quad k = (1, \ldots, K) \\ \text{with the constraints} \quad \mathbf{U}^\intercal\mathbf{U} = \mathbf{I}_R \quad \text{and} \quad \|\mathbf{v}_k^{(1)}\| = \ldots = \|\mathbf{v}_k^{(R)}\| = 1 \quad \text{for} \quad k = (1, \ldots, K), \end{cases} \tag{1}$$

where

- $\mathbf{U} \in \mathcal{M}_{N,R}(\mathbb{R})$ is the standardized and orthogonal common components,
- $\mathbf{V} \in \mathcal{M}_{P,R}(\mathbb{R})$ is the standardized (not necessarily orthogonal) common loadings,
- $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ the standardized block-loadings, $\Sigma_1 \in \mathcal{M}_{R,R}(\mathbb{R})$
- $\Sigma \in \mathcal{M}_{R,R}(\mathbb{R})$ the diagonal scaling matrices with property $\Sigma^2 = \sum_{k=1}^K \Sigma_k^2$ that relate common and block-analyses.

This factorization is *the canonical factorization*. Each multiblock method derives a common component $\mathbf{U}$ using a different optimization criterion. Projecting the data onto these common components enables their canonical factorization (for details see [Bougeard et al., 2024]). The benchmark in [Bougeard et al., 2024] showed that all the methods under study produced similar results. The decomposition of the concatenated block $\mathbf{X}$ is unique for all the multiblock methods, however, only assuming $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$ guarantees uniqueness of the decomposition of each block $\mathbf{X}_k$.

For $R < rank(\mathbf{X})$, the objective is to determine the optimal approximation of the form given in (1) by minimizing an ordinary least square criterion.

2.1 Canonical Factorisation with orthogonal block-loadings

Providing a solution for canonical factorization means minimizing

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{2}$$

with $\mathbf{U}^\intercal.\mathbf{U} = I_R$, $\mathbf{V}_k^\intercal.\mathbf{V}_k = I_R$ for $k = 1, \ldots, K$.

Consider $\mathbf{H}_k = \Sigma_k.\mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1|\ldots|\mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. An iterative algorithm for building the solution of (2) requires to update successively $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ as follows:

1. Updating of $\mathbf{U}$: For given $\mathbf{X} \in \mathcal{M}_{N,P}(\mathbb{R})$, $\mathbf{H} \in \mathcal{M}_{R,P}(\mathbb{R})$, update $\mathbf{U}$ by $\mathbf{U} = \mathbf{A}.\mathbf{B}^\intercal$ where $\mathbf{A}, \mathbf{B}$ are respectively the matrices of the first R left singular vectors and the matrix of the first R right singular vectors from the svd of $\mathbf{X}.\mathbf{H}^\intercal$.
2. Updating of $\mathbf{V}_k$: For given $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, $\mathbf{U} \in \mathcal{M}_{R,N}(\mathbb{R})$ with orthogonal columns, $\Sigma_k \in \mathcal{M}_{R,R}(\mathbb{R})$, update $\mathbf{V}_k$ by: $\mathbf{V}_k = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}, \mathbf{B}$ are respectively the matrix of the first R left singular vectors and the matrix of the first R right singular vectors from the svd of $\mathbf{X}_k^\intercal.\mathbf{U}$.

3. Updating of Σ_k: For given $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, $\mathbf{U} \in \mathcal{M}_{R,N}(\mathbb{R})$ orthogonal, $\mathbf{V}_k \in \mathcal{M}_{R,p_k}(\mathbb{R})$ with orthonormal columns, Σ_k is given by

$$\Sigma_k = diag(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k).$$

Proposition 1. *The successive updates of* $\mathbf{U}$, $\mathbf{V}_k$, *and* Σ_k *guarantee that the criterion defined in* (2) *decreases or remains constant at each step, ensuring convergence of the algorithm.*

3 Bayesian framework for Canonical Factorisation

A probabilistic counterpart of (2) is obtained by assuming a Gaussian noise:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \text{ with i.i.d. entries } \mathcal{N}\big(0, \tau_k^{-1}\big), \qquad k = 1, \dots, K, \tag{3}$$

where τ_k is the noise precision of block k. Let $\mathcal{V}_{m,H} = \{\mathbf{W} \in \mathcal{M}_{m,H}(\mathbb{R}) : \mathbf{W}^\intercal\mathbf{W} = \mathbf{I}_H\}$ denote the Stiefel manifold. Since the blocks are centered, the latent components are constrained to be centered as well: $\mathbf{U} \in \mathcal{V}_{N,H}$ with $\mathbf{1}_N^\intercal\mathbf{U} = 0$.

Priors. We use the following independent priors:

- $\mathbf{U}$ and $\mathbf{V}_1, \dots, \mathbf{V}_K$ are uniformly distributed on their Stiefel manifolds (non-informative);
- $\sigma_k^{(h)} \sim \mathcal{N}^+\big(0, \lambda^{-1}\big)$, a half-normal distribution with a small precision λ (e.g. $\lambda = 10^{-2}$); the positivity constraint removes the sign indeterminacy between $\sigma_k^{(h)}$ and $\mathbf{v}_k^{(h)}$;
- $\tau_k \sim \text{Gamma}(a_0, b_0)$ with small hyperparameters (e.g. $a_0 = b_0 = 10^{-3}$).

3.1 Full conditional distributions

Likelihood. Because $\mathbf{U}$ and $\mathbf{V}_k$ have orthonormal columns and Σ_k is diagonal,

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\,\text{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}) + \sum_{h=1}^{H}(\sigma_k^{(h)})^2. \tag{4}$$

Since the blocks are centered, each $\mathbf{X}_k$ carries $(N-1)p_k$ degrees of freedom, and the log-likelihood reads

$$\log p(\mathbf{X}_1, \dots, \mathbf{X}_K \mid \cdot) = \sum_{k=1}^{K}\Big[\tfrac{(N-1)p_k}{2}\log\tau_k - \tfrac{\tau_k}{2}\big\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\big\|^2\Big] + \text{const.}$$

By (4), $\mathbf{U}$ and $\mathbf{V}_k$ only appear in the linear term $\text{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U})$. This is the key property: all the full conditionals belong to standard families.

Matrix von Mises–Fisher distribution. A random matrix $\mathbf{W} \in \mathcal{V}_{m,H}$ follows the matrix von Mises–Fisher distribution vMF($\mathbf{F}$), with parameter $\mathbf{F} \in \mathcal{M}_{m,H}(\mathbb{R})$, if its density with respect to the uniform measure on $\mathcal{V}_{m,H}$ is proportional to $\exp\{\text{trace}(\mathbf{F}^\intercal\mathbf{W})\}$ [**?**]. If $\mathbf{F} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the (thin) SVD of $\mathbf{F}$, the mode of vMF($\mathbf{F}$) is the polar factor $\mathbf{A}\mathbf{B}^\intercal$, and the distribution concentrates around it as the singular values of $\mathbf{F}$ increase.

Proposition 2 (Full conditionals)**.** *Under model* (3) *and the above priors:*

1. ***Common space.*** *Let* $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ *be an orthonormal basis of* $\mathbf{1}_N^{\perp}$ *and write* $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$ *with* $\tilde{\mathbf{U}} \in \mathcal{V}_{N-1,H}$*. Then*

$$\tilde{\mathbf{U}} \mid \cdot \;\sim\; \mathrm{vMF}\big(\mathbf{Q}^{\intercal}\mathbf{F}_{\mathbf{U}}\big), \qquad \mathbf{F}_{\mathbf{U}} = \sum_{k=1}^{K} \tau_k \, \mathbf{X}_k \mathbf{V}_k \Sigma_k.$$

2. ***Block loadings.*** *For* $k = 1, \dots, K$*,*

$$\mathbf{V}_k \mid \cdot \;\sim\; \mathrm{vMF}\big(\mathbf{F}_{\mathbf{V}_k}\big), \qquad \mathbf{F}_{\mathbf{V}_k} = \tau_k \, \mathbf{X}_k^{\intercal}\mathbf{U}\Sigma_k.$$

3. ***Block weights.*** *For* $k = 1, \dots, K$ *and* $h = 1, \dots, H$*, let* $d_k^{(h)} = \mathbf{u}^{(h)\intercal}\mathbf{X}_k\mathbf{v}_k^{(h)}$*. Then*

$$\sigma_k^{(h)} \mid \cdot \;\sim\; \mathcal{N}^{+}\Big(\frac{\tau_k \, d_k^{(h)}}{\tau_k + \lambda}, \; \frac{1}{\tau_k + \lambda}\Big),$$

   a normal distribution truncated to $(0, +\infty)$*.*

4. ***Noise precisions.*** *For* $k = 1, \dots, K$*,*

$$\tau_k \mid \cdot \;\sim\; \mathrm{Gamma}\Big(a_0 + \frac{(N-1)p_k}{2}, \; b_0 + \frac{1}{2}\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^{\intercal}\|^2\Big).$$

4 The Gibbs sampler

Algorithm 1 Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; dimension H; number of iterations T; burn-in T_0; hyperparameters a_0, b_0, λ

Initialization: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, $\mathbf{V}_k$, Σ_k from OMCOA (or MCOA, or at random); $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^{\intercal}\|^2$
for $t = 1, \dots, T$ **do**
  $\mathbf{F}_{\mathbf{U}} \leftarrow \sum_k \tau_k \, \mathbf{X}_k\mathbf{V}_k\Sigma_k$; update $\tilde{\mathbf{U}}$ by one column-wise sweep of $\mathrm{vMF}(\mathbf{Q}^{\intercal}\mathbf{F}_{\mathbf{U}})$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \dots, K$ **do**
    $\mathbf{F}_{\mathbf{V}_k} \leftarrow \tau_k \, \mathbf{X}_k^{\intercal}\mathbf{U}\Sigma_k$; update $\mathbf{V}_k$ by one column-wise sweep of $\mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k})$
    for $h = 1, \dots, H$ **do**
      $d_k^{(h)} \leftarrow \mathbf{u}^{(h)\intercal}\mathbf{X}_k\mathbf{v}_k^{(h)}$; draw $\sigma_k^{(h)} \sim \mathcal{N}^{+}\big(\tau_k d_k^{(h)}/(\tau_k+\lambda),\, 1/(\tau_k+\lambda)\big)$
    end for
    draw $\tau_k \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^{\intercal}\|^2/2\big)$
  end for
  if $t > T_0$ **then** store $(\mathbf{U}, \mathbf{V}_1, \dots, \mathbf{V}_K, \Sigma_1, \dots, \Sigma_K, \tau_1, \dots, \tau_K)$
  end if
end for
Post-processing: align each stored draw on a reference solution (Section 4)
return the aligned draws

Post-processing: label switching and signs The posterior distribution is invariant under any permutation π of the H dimensions applied jointly to $\mathbf{U}$, $\mathbf{V}_1, \dots, \mathbf{V}_K$ and $\Sigma_1, \dots, \Sigma_K$, and under a joint change of sign of $\mathbf{u}^{(h)}$ and $\mathbf{v}_1^{(h)}, \dots, \mathbf{v}_K^{(h)}$. Posterior summaries are therefore computed after aligning each draw on a reference $\mathbf{U}_{\text{ref}}$ (in practice the OMCOA solution). For each draw, with $\mathbf{C} = \mathbf{U}_{\text{ref}}^\intercal \mathbf{U}$,

$$\hat{\pi} = \arg\max_{\pi \in \mathfrak{S}_H} \sum_{h=1}^{H} |C_{h,\pi(h)}|, \qquad s_h = \operatorname{sign}\big(C_{h,\hat{\pi}(h)}\big),$$

and the columns of $\mathbf{U}$ and $\mathbf{V}_k$ are reordered by $\hat{\pi}$ and multiplied by s_h, while the diagonal of Σ_k is reordered by $\hat{\pi}$ (it stays positive). Posterior means and credible intervals of $\sigma_k^{(h)}$ are then obtained from the empirical distribution of the aligned draws.

Remark 1 (OMCOA as iterated conditional modes)**.** *The modes of the full conditionals of Proposition 2 are*

$$\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal \text{ with } \mathbf{F}_{\mathbf{U}} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal, \qquad \mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal \text{ with } \mathbf{F}_{\mathbf{V}_k} = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal, \qquad \sigma_k^{(h)} = \frac{\tau_k}{\tau_k + \lambda}\, d_k^{(h)}.$$

When $\lambda \to 0$ and the precisions are equal ($\tau_1 = \cdots = \tau_K$), these are exactly the OMCOA updates: Procrustes rotation of $\mathbf{X}\mathbf{H}^\intercal$ for $\mathbf{U}$ (with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$), Procrustes rotation of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ for $\mathbf{V}_k$, and $\Sigma_k = \operatorname{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. OMCOA is therefore the iterated conditional modes (ICM) algorithm [?] associated with this posterior, and the Gibbs sampler is its stochastic counterpart.

Remark 2 (Block weighting)**.** *With block-specific precisions, $\mathbf{F}_{\mathbf{U}} = \sum_k \tau_k \mathbf{X}_k\mathbf{V}_k\Sigma_k$ weights each block by its estimated noise precision, so that noisier blocks contribute less to the common space. To reproduce exactly the OMCOA criterion (??), a common precision $\tau_1 = \cdots = \tau_K = \tau$ can be used, with $\tau \mid \cdot \sim \text{Gamma}\big(a_0 + \sum_k (N-1)p_k/2,\ b_0 + \sum_k \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.*

Remark 3 (What the sampler adds)**.** *Besides point estimates close to the OMCOA solution, the sampler provides the posterior distribution of the block weights $\sigma_k^{(h)}$ and of the noise levels τ_k^{-1}. Credible intervals on $\sigma_k^{(h)}$ can be used to assess whether a dimension is common to all blocks, shared by some of them, or specific to one block.*

Remark 4 (Practical aspects)**.** *Starting from the OMCOA solution shortens the burn-in. The column-wise sweeps make successive draws of $\mathbf{U}$ and $\mathbf{V}_k$ correlated, so convergence should be checked on trace plots of the $\sigma_k^{(h)}$. Because the posterior is strongly concentrated when the noise is small, the chain rarely jumps between the $H!$ equivalent modes; the alignment step remains necessary nonetheless, in particular when several dimensions have close weights.*

References

[Bougeard et al., 2024] Bougeard, S., Peltier, C., Jaillais, B., Boulet, J.-C., and Hanafi, M. (2024). Benchmarking multiblock methods with canonical factorization. *Chemometrics and Intelligent Laboratory Systems*, 254:105240.

[Carroll, 1968] Carroll, J. D. (1968). Generalization of canonical correlation analysis to three of more sets of variables. In *APA 76th Annual Convention, San Francisco, CA, August 30-September 3, 1968*.

[Chessel and Hanafi, 1996] Chessel, D. and Hanafi, M. (1996). Analyses de la co-inertie de k nuages de points. *Revue de statistique appliquée*, 44(2):35–60.

[Cliff, 1966] Cliff, N. (1966). Orthogonal rotation to congruence. *Psychometrika*, 31(1):33–42.

[Escofier and Pages, 1994] Escofier, B. and Pages, J. (1994). Multiple factor analysis (afmult package). *Computational statistics & data analysis*, 18(1):121–140.

[Horst, 1961] Horst, P. (1961). Relations among m sets of variables. *Psychometrika*, (26):129–149.

[Hotelling, 1935] Hotelling, H. (1935). The most predictable criterion. *Journal of educational Psychology*, 26(2):139.

[Kettenring, 1971] Kettenring, J. R. (1971). Canonical analysis of several sets of variables. *Biometrika*, 58(3):433–451.

[Lavit et al., 1994] Lavit, C., Escoufier, Y., Sabatier, R., and Traissac, P. (1994). The act (statis method). *Computational Statistics & Data Analysis*, 18(1):97–119.

[Qannari et al., 2001] Qannari, E. M., Courcoux, P., and Vigneau, E. (2001). Common components and specific weights analysis performed on preference data. *Food Quality and Preference*, 12(5-7):365–368.

[Westerhuis et al., 1998] Westerhuis, J. A., Kourti, T., and MacGregor, J. F. (1998). Analysis of multiblock and hierarchical pca and pls models. *Journal of Chemometrics: A Journal of the Chemometrics Society*, 12(5):301–321.

[Wold et al., 1996] Wold, S., Kettaneh, N., and Tjessem, K. (1996). Hierarchical multiblock pls and pc models for easier model interpretation and as an alternative to variable selection. *Journal of chemometrics*, 10(5-6):463–482.

Appendix: proof of the proposition

Proof. The criterion to be minimized is:

$$\mathcal{L}(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K) = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\top\|^2,$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$ for all $k = 1, \ldots, K$.

The algorithm is of the *Alternating Least Squares* (ALS) type: at each iteration, the criterion is minimized successively with respect to each parameter, while the others are held fixed. It therefore suffices to show that each individual update decreases or maintains $\mathcal{L}$.

Update of U With $\mathbf{V}_k$ and Σ_k fixed, let $\mathbf{H}_k = \Sigma_k \mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1 \mid \cdots \mid \mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. Then:

$$\mathcal{L} = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\mathbf{H}_k\|^2 = \|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2.$$

Expanding: $\|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2 = \|\mathbf{X}\|^2 - 2\ \mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) + \|\mathbf{U}\mathbf{H}^\intercal\|^2$.

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$: $\|\mathbf{U}\mathbf{H}^\intercal\|^2 = \mathrm{trace}(\mathbf{H}\mathbf{U}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}\mathbf{H}^\intercal) = \|\mathbf{H}\|^2$ (constant w.r.t. $\mathbf{U}$).

Minimizing $\mathcal{L}$ with respect to $\mathbf{U}$ is therefore equivalent to maximizing

$$\text{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \text{trace}(\mathbf{H}^\intercal\mathbf{X}^\intercal\mathbf{U}) = \text{trace}(\mathbf{U}^\intercal\mathbf{X}\mathbf{H}),$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{AB})$ for $\mathbf{A}, \mathbf{B}$ of dimension N, P and P, N. The second equality comes from the equality of the trace of a square matrix and its transpose. By [Cliff, 1966], the maximum is attained at $\mathbf{U}^* = AB^\intercal$, where A (resp. B) are the matrices of the R leading left (resp. right) singular vectors of $\mathbf{X}\mathbf{H}^T$. This update is the global minimizer of $\mathcal{L}$ with respect to $\mathbf{U}$, hence:

$$\mathcal{L}(\mathbf{U}^*, \ldots) \leq \mathcal{L}(\mathbf{U}_{\text{old}}, \ldots).$$

Update of $\mathbf{V}_k$ With $\mathbf{U}$ and Σ_k fixed, expanding the contribution of block k:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \text{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) + \|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2.$$

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$:

$$\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \text{trace}(\mathbf{V}_k\Sigma_k^\intercal\mathbf{U}^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \text{trace}(\Sigma_k^2) \quad \text{(constant w.r.t. } \mathbf{V}_k\text{)}.$$

Minimizing therefore reduces to maximizing $\text{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \text{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k)$ subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$. Since the diagonal matrix $\Sigma_k > 0$ does not affect the singular vectors, the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}$, $\mathbf{B}$ are obtained from the SVD of $\mathbf{X}_k^\intercal\mathbf{U}$ [Cliff, 1966] . Hence:

$$\mathcal{L}(\ldots, \mathbf{V}_k^*, \ldots) \leq \mathcal{L}(\ldots, \mathbf{V}_{k,\text{old}}, \ldots).$$

Update of Σ_k With $\mathbf{U}$ and $\mathbf{V}_k$ fixed (both having orthonormal columns), and $\Sigma_k = \text{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, the contribution of block k is:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \text{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k) + \text{trace}(\Sigma_k^2).$$

Since Σ_k is diagonal, it remains to minimazing:

$$\sum_{r=1}^{R}\Big[()\sigma_k^{(r)})^2 - 2\,(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}\cdot d_r\Big].$$

Each term is a convex parabola in $\sigma_k^{(r)}$, minimized at: $(\sigma_k^{(r)})^* = (\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}$, or, in matrix form:$\Sigma_k^* = \text{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. This update is therefore the exact minimizer with respect to Σ_k, and:

$$\mathcal{L}(\ldots, \Sigma_k^*, \ldots) \leq \mathcal{L}(\ldots, \Sigma_{k,\text{old}}, \ldots).$$

Global Convergence At each complete iteration (update of $\mathbf{U}$, then all $\mathbf{V}_k$, then all Σ_k), the criterion $\mathcal{L}$ is monotonically decreasing and bounded below by zero. By the monotone convergence theorem, the sequence of values $(\mathcal{L}^{(t)})_{t\geq 0}$ converges:

$$\mathcal{L}^{(t+1)} \leq \mathcal{L}^{(t)}, \quad \mathcal{L}^{(t)} \geq 0 \implies \mathcal{L}^{(t)} \xrightarrow[t\to\infty]{} \ell^* \geq 0.$$

□

Sketch of proof. (1)–(2): by (4), the conditional density of $\mathbf{U}$ is proportional to $\exp\{\sum_k \tau_k \operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})\} = \exp\{\operatorname{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{U})\}$, and $\operatorname{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{Q}\tilde{\mathbf{U}}) = \operatorname{trace}\big((\mathbf{Q}^\intercal \mathbf{F}_\mathbf{U})^\intercal \tilde{\mathbf{U}}\big)$; the same argument applies to $\mathbf{V}_k$ with $\operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U}) = \operatorname{trace}(\mathbf{F}_{\mathbf{V}_k}^\intercal \mathbf{V}_k)/\tau_k$. (3): the terms of (4) involving $\sigma_k^{(h)}$ are $-\frac{\tau_k}{2}\big[(\sigma_k^{(h)})^2 - 2\sigma_k^{(h)} d_k^{(h)}\big]$; adding the log-prior $-\frac{\lambda}{2}(\sigma_k^{(h)})^2$ and completing the square gives a Gaussian kernel, restricted to $\sigma_k^{(h)} > 0$. (4): the Gamma prior is conjugate for the precision of a Gaussian likelihood. □

Sampling from the matrix von Mises–Fisher distribution Direct sampling from $\mathrm{vMF}(\mathbf{F})$ is not straightforward, but its column-wise conditionals are vector von Mises–Fisher distributions [**?**]. Let $\mathbf{W}_{-j}$ be $\mathbf{W}$ without its j-th column and $\mathbf{N}_j \in \mathcal{M}_{m,m-H+1}(\mathbb{R})$ an orthonormal basis of $\operatorname{span}(\mathbf{W}_{-j})^\perp$. Then

$$\mathbf{w}_j = \mathbf{N}_j \mathbf{z}, \qquad \mathbf{z} \sim \mathrm{vMF}\big(\mathbf{N}_j^\intercal \mathbf{f}_j\big) \text{ on the unit sphere } \mathbb{S}^{m-H},$$

where $\mathbf{f}_j$ is the j-th column of $\mathbf{F}$. One sweep over the H columns (in random order) is a valid Gibbs update of $\mathbf{W}$, which is enough inside the Gibbs sampler. The vector distribution $\mathrm{vMF}(\mathbf{f})$, with density proportional to $\exp(\mathbf{f}^\intercal \mathbf{z})$ on $\mathbb{S}^{m-1}$, is sampled with the rejection algorithm of Wood [**?**] (Algorithm 2).

Algorithm 2 Sampling $\mathbf{z} \sim \mathrm{vMF}(\mathbf{f})$ on $\mathbb{S}^{m-1}$ (Wood, 1994)

$\kappa \leftarrow \|\mathbf{f}\|, \quad \boldsymbol{\mu} \leftarrow \mathbf{f}/\kappa$
$b \leftarrow (m-1)/\big(2\kappa + \sqrt{4\kappa^2 + (m-1)^2}\big), \quad x_0 \leftarrow (1-b)/(1+b), \quad c \leftarrow \kappa x_0 + (m-1)\log(1-x_0^2)$
repeat
    draw $\zeta \sim \mathrm{Beta}\big(\frac{m-1}{2}, \frac{m-1}{2}\big)$ and $\upsilon \sim \mathcal{U}(0,1)$
    $w \leftarrow \big(1-(1+b)\zeta\big)/\big(1-(1-b)\zeta\big)$
until $\kappa w + (m-1)\log(1-x_0 w) - c \geq \log \upsilon$
draw $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_m)$, set $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} - (\boldsymbol{\mu}^\intercal \boldsymbol{\xi})\boldsymbol{\mu}$ and $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi}/\|\boldsymbol{\xi}\|$
return $\mathbf{z} = w\,\boldsymbol{\mu} + \sqrt{1-w^2}\,\boldsymbol{\xi}$

The expression of b used in Algorithm 2 is algebraically equivalent to Wood's original one, $b = (-2\kappa + \sqrt{4\kappa^2 + (m-1)^2})/(m-1)$, but avoids a numerical cancellation when κ is large, which is the typical situation here since κ scales with the noise precisions τ_k.